

Emmy STACK01

emmystack01.com

Website — Full Build Summary

A complete record of every system built, debugged, and deployed across the Emmy STACK01 agency website — from navigation to CMS, SEO to legal documents.

emmystack01.com | [@Emmy_STACK01](https://twitter.com/Emmy_STACK01)

Project Overview

Emmy STACK01 (emmystack01.com) is a Nigerian digital agency website offering eight website build types, a Digital Business Card product, per-page custom project pricing, a free tools suite, an affiliate/partnership program with lifetime client attribution, and a blog called 'The Digital DNA Log'. The agency operates on a pay-after-site-is-ready trust model and communicates primarily through WhatsApp and X (@Emmy_STACK01).

Tech Stack

- **Framework:** Astro ^6.4.4 — static-first with server output for CMS routes
- **Deployment:** Vercel (adapter: @astrojs/vercel ^10.0.8)
- **CMS:** Keystatic @5.1.0 / @keystatic/core @0.5.50 — GitHub storage mode
- **Styling:** Custom CSS with CSS variables, dark glass aesthetic
- **Fonts:** Orbitron (headings), Montserrat (body)
- **Accent colour:** Cyan (#00ccff)
- **React:** ^18.3.1 (required by Keystatic)
- **Repo:** EmmyStack01/EmmySTACK01 (GitHub, branch: main)
- **Node:** 22.x

1. Site Structure & Pages

The site is built in Astro with output: 'server' to support Keystatic's API routes. Pages are organized under src/pages/ following Astro's file-based routing.

Key pages and routes

- **/** — Homepage with hero, services overview, CTA
- **/products** — Products listing page
- **/products/digital-business-card** — DBC product page
- **/tools/brand-engine** — Brand engine free tool
- **/tools/website-auditor** — Website auditor tool (Digital DNA Auditor)
- **/tools/emmy-sign** — Emmy Sign tool
- **/tools/naira-wise** — Naira Wise tool
- **/tools/social-architect** — Social Architect tool
- **/tools/meeting_background_studio** — Meeting background tool
- **/partnership/affiliate** — Affiliate program page
- **/partnership/affiliate/partner-kit** — Partner kit download
- **/partnership/policy** — Policy page

- **/articles** — Blog index ('The Digital DNA Log')
- **/articles/[slug]** — Individual article pages
- **/keystatic** — Keystatic CMS admin (protected)

Short URL redirects (vercel.json)

```

/partner-kit      → /partnership/affiliate/partner-kit
/affiliate        → /partnership/affiliate
/product          → /products
/brand-engine     → /tools/brand-engine
/website-auditor  → /tools/website-auditor
/policy           → /partnership/policy
/emmy-sign        → /tools/emmy-sign
/naira-wise       → /tools/naira-wise
/social-architect → /tools/social-architect
/digital-business-card → /products/digital-business-card
/meeting_background_studio → /tools/meeting_background_studio
/document-engine  → rewrites to GitHub Pages (Emmy Document Engine)

```

2. Navigation System

The navbar and mobile drawer were built and debugged across multiple sessions. The system uses a dark glass aesthetic with CSS variables, Orbitron/Montserrat fonts, and cyan (#00ccff) accents.

Files

- **index-style.css** — main CSS file containing all nav/drawer styles
- **navbar.js** — desktop navbar behaviour (dropdowns, active states)
- **nav-link.js** — handles active nav link highlighting per page

Key bugs fixed

- Incorrect HTML nesting — overlay and drawer elements were inside <nav> instead of being siblings at body level
- Missing id='openMenu' on the .nav-toggle button — JS could not find the hamburger trigger
- Missing inline script block that initialized the drawer on page load
- Dropdown race condition — fixed by replacing .classList.toggle() with an explicit isOpen flag and separate open/close logic

Delivery preferences established

- Files always delivered as downloads (not inline in chat)
- Existing class/id names preserved during integration
- Both HTML and CSS files updated together when changes span both

3. Keystatic CMS Integration

Keystatic CMS was integrated to power 'The Digital DNA Log' (articles/blog). It uses GitHub storage mode — edits made in the Keystatic dashboard commit directly to the EmmySTACK01/EmmySTACK01 GitHub repo.

Storage configuration

```

// keystatic.config.ts
storage: {
  kind: 'github',
  repo: 'EmmyStack01/EmmySTACK01',
  clientId: process.env.KEYSTATIC_GITHUB_CLIENT_ID,
  clientSecret: process.env.KEYSTATIC_GITHUB_CLIENT_SECRET,
}

```

```
// KEYSTATIC_SECRET is read from process.env automatically
```

Articles collection schema

- **title** — fields.slug (generates URL slug from title)
- **author** — fields.text (default: 'Wisdom Ezuduemoih')
- **description** — fields.text multiline (SEO hook/preview)
- **date** — fields.date (default: today)
- **category** — fields.select (Branding, Web Development, Technical SEO, Partnership)
- **readingTime** — fields.text (e.g. '5 min read')
- **coverImage** — fields.image (saves to public/asset/, publicPath: /asset/)
- **content** — fields.document with formatting, dividers, links, inline images, and custom component blocks

Custom component blocks in document field

- **cta** — Contextual Action Card (title, description text, button text, button link)

Singletons

- **settings** — Site Settings singleton (siteTitle, global SEO description, WhatsApp CTA link)

Article content rendering

A DocumentRenderer component was extracted into a separate ArticleRenderer.tsx file to handle JSX rendering inside Astro's [slug].astro page. This was required because Astro .astro files cannot render Keystatic's document field output directly.

Version compatibility — critical

Astro 6 introduced breaking changes that required specific Keystatic versions:

```
"@keystatic/astro": "^5.1.0"    (not 5.0.0 — routes not injected in Astro 6)
"@keystatic/core": "^0.5.50"   (not 0.5.0 — missing GitHub refresh_token support)
```

Bugs fixed during integration

- @keystatic/astro version 5.0.0 + Astro 6: integration did not inject /api/keystatic/[...params] route — resolved by upgrading to 5.1.0
- react-dom/client export error: caused by Vite caching stale build — resolved by clearing .astro/ cache and node_modules
- react-aria version mismatch: downgraded to compatible version
- Component blocks (Alert/Tip Box, Contextual Action Card) not appearing in editor: fixed by switching from fields.markdoc() to fields.document() with componentBlocks
- DocumentRenderer JSX error in .astro file: fixed by extracting into separate ArticleRenderer.tsx component

4. Vercel Deployment

The site is deployed on Vercel using the @astrojs/vercel adapter. Vercel was chosen over Cloudflare because Keystatic's GitHub storage mode integrates without issues on Vercel's SSR infrastructure.

astro.config.mjs — key settings

```
export default defineConfig({
  site: 'https://emmystack01.com',
  trailingSlash: 'ignore',
  output: 'server',
  adapter: vercel({ webAnalytics: { enabled: true } }),
  legacy: { collectionsBackwardsCompat: true },
```

```

integrations: [react(), markdoc(), keystatic(), sitemap(), indexnow()],
vite: {
  build: { sourcemap: false },
  resolve: { dedupe: ['react', 'react-dom'] },
  optimizeDeps: { include: ['react', 'react-dom', 'react-dom/client'] },
},
});

```

Vercel environment variables

- **KEYSTATIC_GITHUB_CLIENT_ID** — GitHub App Client ID
- **KEYSTATIC_GITHUB_CLIENT_SECRET** — GitHub App Client Secret
- **KEYSTATIC_SECRET** — minimum 32-character random string

All three set for All Environments (Production, Preview, Development).

vercel.json configuration

- **installCommand**: npm install --legacy-peer-deps (handles peer dep conflicts)
- **cleanUrls**: true (removes .html extensions from URLs)
- **redirects**: short URL aliases (see Section 1)
- **rewrites**: /document-engine proxies to GitHub Pages
- **headers**: CORS headers on /api/*, COOP/COEP on all routes

Key deployment issue — Keystatic route not generated

On Astro 6, Keystatic's @keystatic/astro integration injected the /api/keystatic/[...params] route but it did not appear as a separate Vercel function. Investigation revealed Astro 6 bundles all SSR routes into a single /_render function — the route existed but wasn't visible in the Vercel Resources tab. The fix was adding a Vercel rewrite in vercel.json to ensure /api/keystatic/* traffic reached /_render:

```

{
  "source": "/api/keystatic/:path*",
  "destination": "/_render"
}

```

5. GitHub App for Keystatic Authentication

Keystatic's GitHub storage mode requires a GitHub App (not an OAuth App). Standard OAuth Apps do not return a refresh_token, which Keystatic requires for session management. Using an OAuth App causes 'Authorization failed' at the OAuth callback stage.

GitHub App settings

- **Name**: EmmySTACK01 (or similar)
- **Homepage URL**: https://emmystack01.com
- **Callback URL**: https://emmystack01.com/api/keystatic/github/oauth/callback
- **Webhook**: disabled (Active unchecked)
- **Permissions**: Repository > Contents: Read & Write, Metadata: Read-only
- **Installation**: Only on EmmyStack01/EmmySTACK01 repo

Bugs fixed during GitHub App setup

- 'Authorization failed': caused by using OAuth App instead of GitHub App — OAuth App missing refresh_token in response
- 404 on /api/keystatic/github/login: route existed inside /_render but needed explicit Vercel rewrite to reach it
- COOP header blocking OAuth redirect: Cross-Origin-Opener-Policy: same-origin on all routes interfered with the GitHub redirect flow — fixed by overriding to 'unsafe-none' specifically for /api/keystatic/* routes

- KEYSTATIC_SECRET too short: must be minimum 32 characters

6. Emmy STACK01 Document Engine

The Document Engine is a single-file HTML tool that generates styled A4 legal document previews with PDF export. It is hosted separately on GitHub Pages (emmystack01.github.io/Emmy_STACK01-Document-Engine-Website/) and proxied through emmystack01.com/document-engine via a Vercel rewrite.

Document types supported

- White-Label License Agreement
- Project Development Agreement

Tech used in the Document Engine

- **Fonts:** Orbitron (headings), Montserrat (body)
- **PDF export:** jsPDF + html2canvas
- **Styling:** A4-sized preview with Emmy STACK01 branding

Bugs fixed

- Incorrect font references: 'Syne' and 'DM Sans' remnants replaced with correct Orbitron/Montserrat references
- hyphens: auto on preview causing on-screen word breaks — removed from screen styles, kept only for print/PDF
- Misconfigured PDF sandbox measurement element causing layout shifts in the exported PDF
- Missing word-merge fix in the onclone PDF export callback — words were splitting incorrectly across lines in the exported PDF

vercel.json rewrite

```
{
  "source": "/document-engine",
  "destination": "https://emmystack01.github.io/Emmy_STACK01-Document-Engine-Website/"
},
{
  "source": "/document-engine/:path*",
  "destination": "https://emmystack01.github.io/Emmy_STACK01-Document-Engine-Website/:path*"
}
```

7. Digital DNA Auditor Tool

The Digital DNA Auditor is a Lighthouse-based website auditing tool available at </tools/website-auditor>. It uses GSAP for animations and Vanta.js for the background effect. Users enter a URL and receive a performance/SEO audit with a downloadable PDF report.

Bugs fixed — blank PDF download

The PDF download was generating a blank/empty file. Three root causes were identified:

- **z-index: -999 on #pdf-template:** The hidden PDF template element had a negative z-index which removed it from the browser's compositing layer — html2canvas could not capture it. Fixed with `visibility: hidden + position: absolute + left: -100vw` instead of z-index tricks
- **display: block override:** A CSS override on the template container broke its internal layout — fixed by removing the conflicting rule
- **Insufficient font-load delay:** html2canvas was capturing the element before custom fonts (Orbitron/Montserrat) had loaded — fixed by awaiting `document.fonts.ready` then adding a 300ms safety delay before capture

8. Pricing Page CSS Fix

A CSS debugging session on pricing.html resolved .footer-socials a links rendering white/invisible on the pricing page.

Root cause

An overly broad wildcard CSS rule was overriding all descendant element colours:

```
.info-box * { color: #ffffff !important; }
```

This !important wildcard was overriding the social link colour styles, making them invisible against the white background of that section.

Fix

Scoped the wildcard to explicit element selectors instead of *, so only intended elements received the white colour override:

```
.info-box h2,  
.info-box h3,  
.info-box p,  
.info-box span {  
  color: #ffffff !important;  
}  
/* .footer-socials a left unaffected – uses its own colour rule */
```

9. SEO — Digital Business Card Prototype Pages

Comprehensive SEO enrichment was applied to multiple Digital Business Card prototype HTML files under the Emmy STACK01 brand, targeting Google Knowledge Panels, Rich Results, and Google Autocomplete signals.

SEO elements added to each card

- Schema.org JSON-LD @graph structured data with profession-specific types beyond generic Person (e.g. Physician for Dr. Amara Okafor, SportsOrganization for Chloe Sterling)
- Inline Microdata on HTML elements (itemprop attributes)
- Open Graph meta tags (og:type, og:title, og:description, og:image, og:url)
- Twitter Card meta tags (twitter:card, twitter:site, twitter:creator)
- Canonical URL tags
- ARIA roles on interactive elements
- Preload directives for critical fonts and images
- Enriched vCard downloads with Schema.org-aligned contact data

Cards completed

- **Alex Rivera** — UI/UX Designer
- **Aisha Patel** — Fashion Designer
- **Dr. Amara Okafor** — Pediatrician
- **Chloe Sterling** — Tennis Content Creator

Cards not yet processed

- Jordan Banks
- Kofi Mensah
- Mily Wayz (handled separately as a full CMS-powered DBC)

10. Mily Wayz DBC — Vercel Rewrite (Pending)

The Mily Wayz Digital Business Card is deployed separately on Cloudflare Workers at <https://milywayz.wezuduemoih.workers.dev>. A planned Vercel rewrite in `emmystack01.com`'s `vercel.json` will alias it under the main domain:

```
{
  "source": "/mily-wayz/:path*",
  "destination": "https://milywayz.wezuduemoih.workers.dev/:path*"
}
```

After adding this rewrite:

- <https://emmystack01.com/mily-wayz/> → serves the Mily Wayz card
- <https://emmystack01.com/mily-wayz/keystatic> → serves the CMS dashboard

Note: After adding the rewrite, update the GitHub App callback URL for Mily Wayz CMS to:
<https://emmystack01.com/mily-wayz/api/keystatic/github/oauth/callback>

11. Other Site Features & Tools

Free Tools Suite

- **Brand Engine** — brand identity generator tool
- **Emmy Sign** — digital signature tool
- **Naira Wise** — Nigerian currency/finance tool
- **Social Architect** — social media planning tool
- **Meeting Background Studio** — virtual background generator
- **Document Engine** — legal document generator (proxied from GitHub Pages)

Affiliate / Partnership Program

- Lifetime client attribution model — affiliates earn commission on all future projects from referred clients, not just the first
- Partner kit download available at </partnership/affiliate/partner-kit>
- Policy page at </partnership/policy>

Blog — The Digital DNA Log

- Powered by Keystatic CMS (GitHub storage mode)
- Articles stored as JSON + document content in `src/content/articles/`
- Content Config (`content.config.ts`) defines the Astro collection schema
- `ArticleRenderer.tsx` handles document field JSX rendering
- Sitemap auto-generated by `@astrojs/sitemap`
- IndexNow submission via `astro-indexnow` on every build

Web Analytics

Vercel Web Analytics enabled via the adapter config:

```
adapter: vercel({ webAnalytics: { enabled: true } })
```

12. Full `vercel.json` (Current State)

```
{
  "version": 2,
  "framework": "astro",
  "buildCommand": "npm run build",
  "installCommand": "npm install --legacy-peer-deps",
```

```

"cleanUrls": true,
"redirects": [
  { "source": "/partner-kit", "destination": "/partnership/affiliate/partner-kit", "permanent": true },
  { "source": "/affiliate", "destination": "/partnership/affiliate", "permanent": true },
  { "source": "/product", "destination": "/products", "permanent": true },
  { "source": "/brand-engine", "destination": "/tools/brand-engine", "permanent": true },
  { "source": "/website-auditor", "destination": "/tools/website-auditor", "permanent": true },
  { "source": "/policy", "destination": "/partnership/policy", "permanent": true },
  { "source": "/emmy-sign", "destination": "/tools/emmy-sign", "permanent": true },
  { "source": "/naira-wise", "destination": "/tools/naira-wise", "permanent": true },
  { "source": "/social-architect", "destination": "/tools/social-architect", "permanent": true },
  { "source": "/digital-business-card", "destination": "/products/digital-business-card", "permanent": true },
  { "source": "/meeting_background_studio", "destination": "/tools/meeting_background_studio", "permanent": true }
],
"rewrites": [
  { "source": "/api/keystatic/:path*", "destination": "/_render" },
  { "source": "/document-engine", "destination": "https://emmystack01.github.io/..." },
  { "source": "/document-engine/:path*", "destination": "https://emmystack01.github.io/.../:path*" }
],
"headers": [
  {
    "source": "/api/(.*)",
    "headers": [
      { "key": "Access-Control-Allow-Credentials", "value": "true" },
      { "key": "Access-Control-Allow-Origin", "value": "*" },
      { "key": "Access-Control-Allow-Methods", "value": "GET,OPTIONS,POST" },
      { "key": "Access-Control-Allow-Headers", "value": "Content-Type" }
    ]
  },
  {
    "source": "/api/keystatic/(.*)",
    "headers": [
      { "key": "Cross-Origin-Opener-Policy", "value": "unsafe-none" }
    ]
  },
  {
    "source": "/(.*)",
    "headers": [
      { "key": "Cross-Origin-Opener-Policy", "value": "same-origin" },
      { "key": "Cross-Origin-Embedder-Policy", "value": "unsafe-none" }
    ]
  }
]
}

```

13. package.json — Key Dependencies

```

{
  "dependencies": {
    "@astrojs/markdoc": "^1.0.6",
    "@astrojs/react": "^4.4.2",
    "@astrojs/sitemap": "^3.2.1",
    "@astrojs/vercel": "^10.0.8",
    "@keystatic/astro": "^5.1.0",
    "@keystatic/core": "^0.5.50",
    "@react-aria/utils": "^3.34.1",
    "@react-stately/utils": "^3.12.1",
    "@types/react": "^18.3.12",
    "@types/react-dom": "^18.3.1",
    "astro": "^6.4.4",
    "astro-indexnow": "^2.1.1",
    "node-fetch": "^3.3.2",
    "react": "^18.3.1",
    "react-aria": "^3.49.0",
    "react-dom": "^18.3.1"
  }
}

```



```
  },  
  "engines": { "node": "22.x" }  
}
```

Note: Install command uses --legacy-peer-deps to resolve react-aria peer conflicts with the React version required by Keystatic.

14. tsconfig.json

```
{  
  "extends": "astro/tsconfigs/base",  
  "compilerOptions": {  
    "jsx": "react-jsx",  
    "jsxImportSource": "react"  
  }  
}
```

jsx and jsxImportSource are required for Keystatic's .tsx config file and the ArticleRenderer.tsx component to compile correctly.

15. Blog Content Edit Flow

Visit <https://emmystack01.com/keystatic>

- Log in with GitHub (via GitHub App)
- Go to Articles (Log) collection
- Create new article or edit existing
- Fill in: title, author, description, date, category, reading time, cover image, content (with component blocks)
- Click Create / Save
- Keystatic commits JSON + content files to EmmyStack01/EmmySTACK01 repo
- Vercel auto-redeploys (triggered by GitHub push)
- New article live at [emmystack01.com/articles/\[slug\]](https://emmystack01.com/articles/[slug])
- Sitemap auto-updated, IndexNow submits new URL to Bing/Google

16. Pending Items

- **Vercel rewrite for Mily Wayz** — add /mily-wayz/:path* rewrite to vercel.json pointing at milywayz.wezuduemoih.workers.dev
- **SEO enrichment for remaining DBC prototypes** — Jordan Banks, Kofi Mensah not yet processed
- **Keystatic Cloud template** — reusable Astro+Keystatic admin project using Keystatic Cloud storage (no GitHub App per client) documented in the DBC Build Template PDF